

**WELCOME
TO
CSE DEPARTMENT**

CONGRATULATIONS!!

Overview of The Course

- Introduction to Computer Systems (CSE 1110)
- Course Credit: 1.0
- Lecturer Abdullah Al Jobair

Objective of the Course

To gain an overview of Computer systems and learn basic computer programming

Introduction to computer

Computer



- ❑ Computer is an electronic device that stores, retrieves, and processes data.
- ❑ It takes data as input, process it and then give information as output.
- ❑ Processed data is called information.
- ❑ Input is processed, produce output – this is directed by the software but performed by the hardware.

Characterstics of Computer

1. Automatic

- ❖ Computer is an automatic machine.
- ❖ Once a program is given to computer i.e. stored in computer memory, the program and instruction can control the program execution without human interaction.

2. High Speed

- ❖ Computer is a very fast device. It is capable of performing calculation of very large amount of data.
- ❖ The computer has units of speed in microsecond, nanosecond, and even the picosecond.
- ❖ It can perform several billions (10^9), even trillions (10^{12}) simple arithmetic calculations in a second

Characterstics of Computer

3. Accuracy

- ❖ The calculations are 100% error free.
- ❖ Computers perform all jobs with 100% accuracy provided that correct input has been given.

4. Diligence

- ❖ Unlike human beings, a computer is free from **monotony, tiredness and lack of concentration**.
- ❖ It can work continuously without any error and boredom.
- ❖ It can do **repeated** work with **same speed** and **accuracy**.

Continue..

Characterstics of Computer

5. Versatility

- ❖ A computer is a very versatile machine.
- ❖ A computer is very flexible in performing the jobs to be done.
- ❖ This machine can be used to solve the problems related to various fields.
- ❖ At one instance, it may be solving a complex scientific problem and the very next moment it may be playing a card game.

6. Power of Remembering

- ❖ Memory is a very important characteristic of computers.
- ❖ A computer has much more storage capacity than human beings.
- ❖ It can store large amount of data.
- ❖ It can store any type of data such as images, videos, text, audio and many others.

Continue..

Characterstics of Computer

7. No I.Q

- ❖ A computer is a machine that has no intelligence to perform any task.
- ❖ Each instruction has to be given to computer.
- ❖ A computer cannot take any decision on its own.

8. No Feeling

- ❖ Computers have no feelings or emotions.
- ❖ It cannot make judgment based on feeling, taste, experience, and knowledge unlike a human being.

Evolution of Computer

First Generation

1946-1959. Vacuum tube based. ENIAC, IBM 701

Second Generation

1959-1965. Transistor based. Honeywell 400, IBM 7030

Third Generation

1965-1971. Integrated Circuit based. IBM 360/370, CDC 6000

Fourth Generation

1971-1980. VLSI microprocessor based. CRAY-1(Super Computer),
CRAY-X-MP(Super Computer), DEC 10

Fifth Generation

1980-onwards. ULSI microprocessor based Desktop, Laptop

Introduction to computer **Types**

1. PC (Personal Computer)

It is a single user computer system having moderately powerful microprocessor



Continue..

2. WorkStation

It is also a single user computer system which is similar to personal computer but have more powerful microprocessor.



Continue..

3. Notebook Computer

Small in size. Used by individual users.



Continue..

4. Tablet Computer

Small in size. Used by individual users.

It has all notebook's functionality but lighter than notebook.

Accept input from a stylus or digital pen.



5. Handheld Computer

Small enough to fit in your hand.

Popular handheld computer is PDA (Personal Digital Assitant).

Not lager than a small appointment book



Continue..

6. Smart Phone

Some cellular phones doubled as miniature PCs
Offers advanced features



In terms of sizes

1. Super Computer

- It is an extremely fast computer which can execute hundreds of millions of instructions per second.
- Most powerful in terms of data processing and performance.



Continue..

2. Main Frame

- Not as powerful as supercomputer
- Mainframes can also process & store large amount of data
- Used in large organization



Continue..

Introduction to computer Types

3. Minicomputer

- Used for specific purposes
- **“Midrange Computers”**
- Individual departments of a large company or organizations



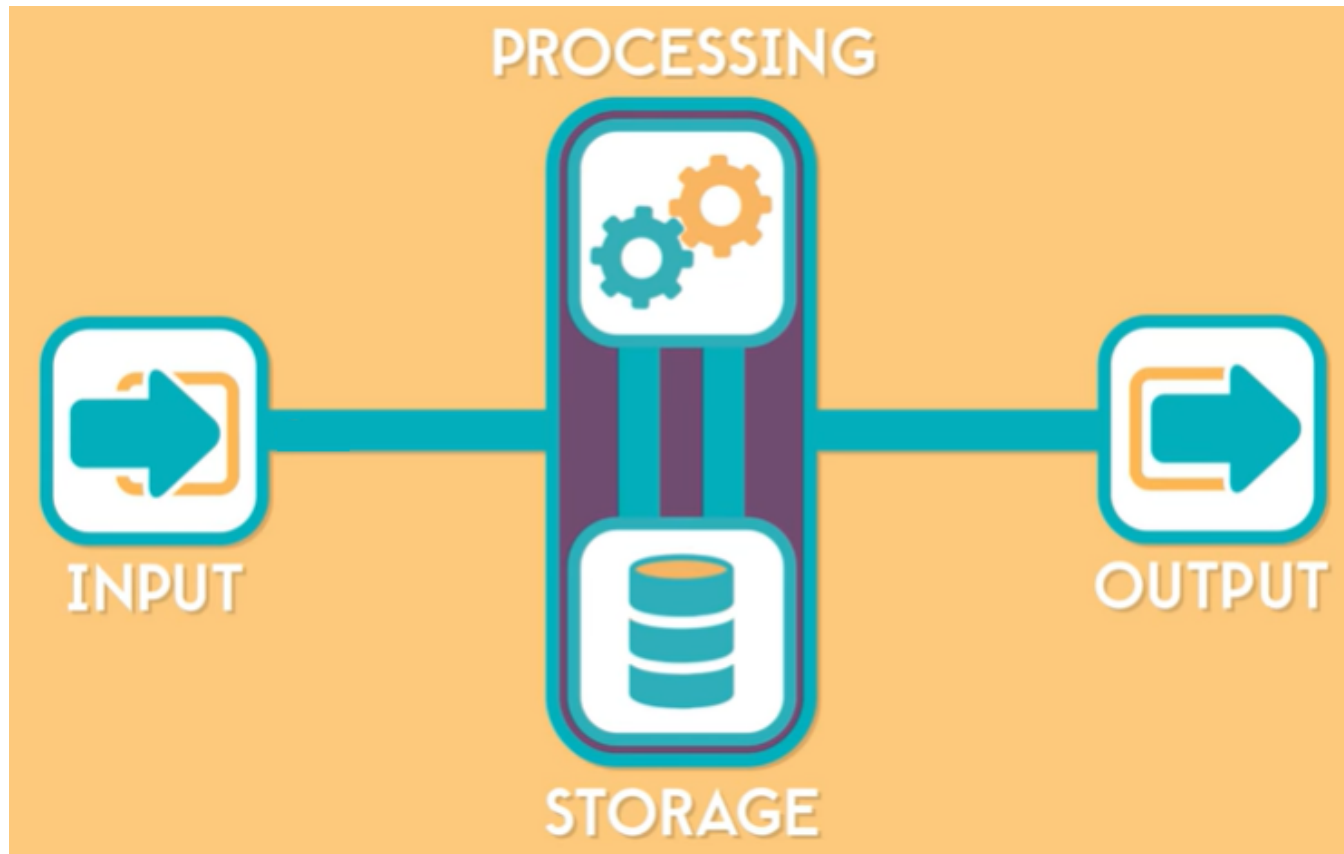
4. Microcomputer

- Desktop , laptops, personal digital assistant (PDA), tablets & smart phones
- Widely used & the fastest growing computers



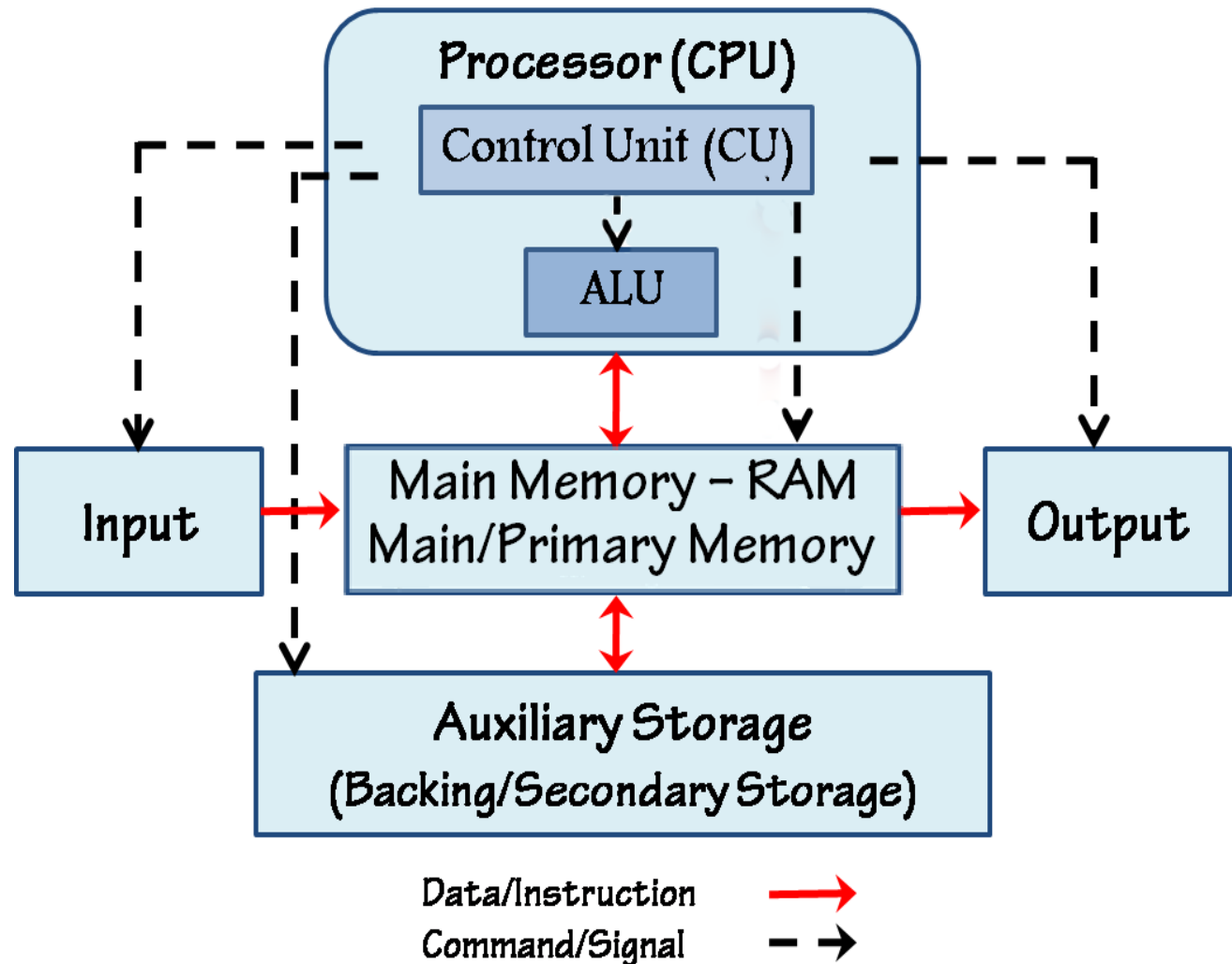
What makes a computer,
a computer?

What makes a computer, a computer?



Basic Functions of Computer

1. Inputting
2. Storing
3. Processing
4. Outputting
5. Controlling

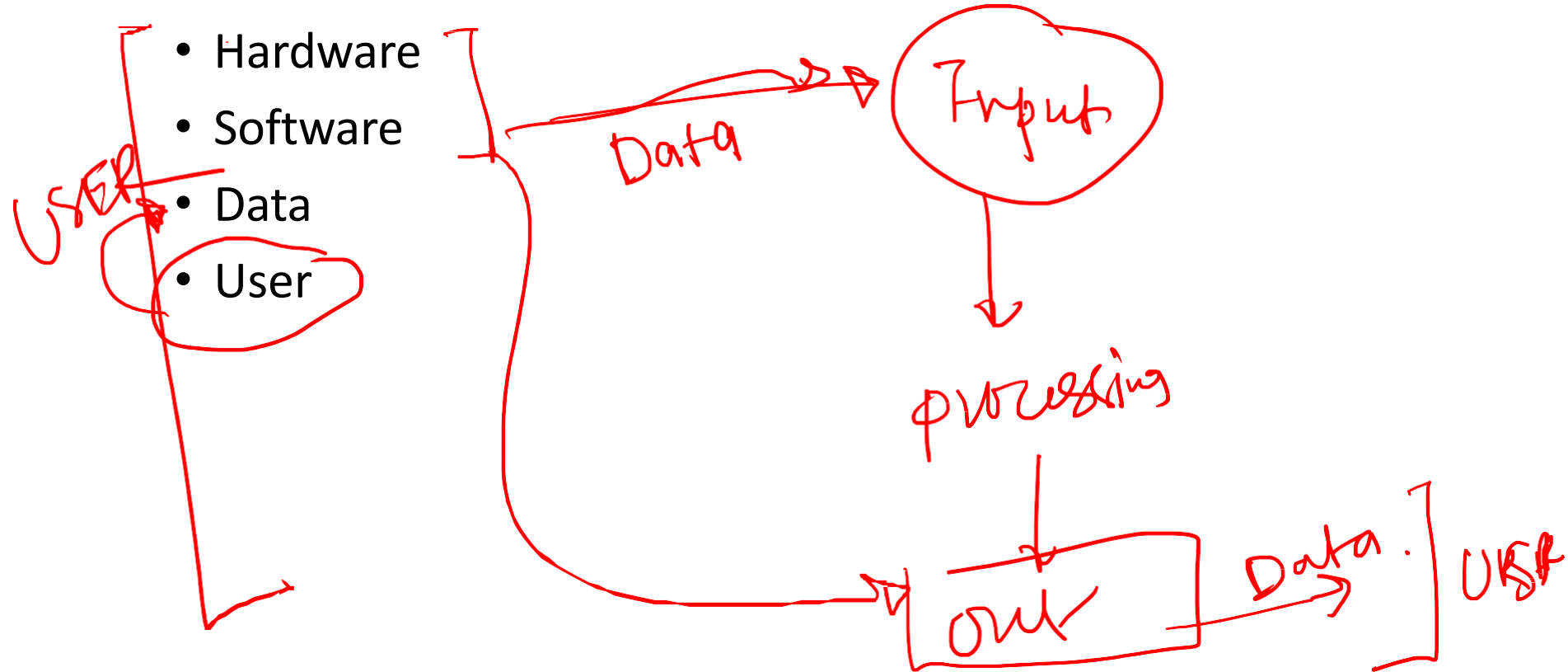


Input >

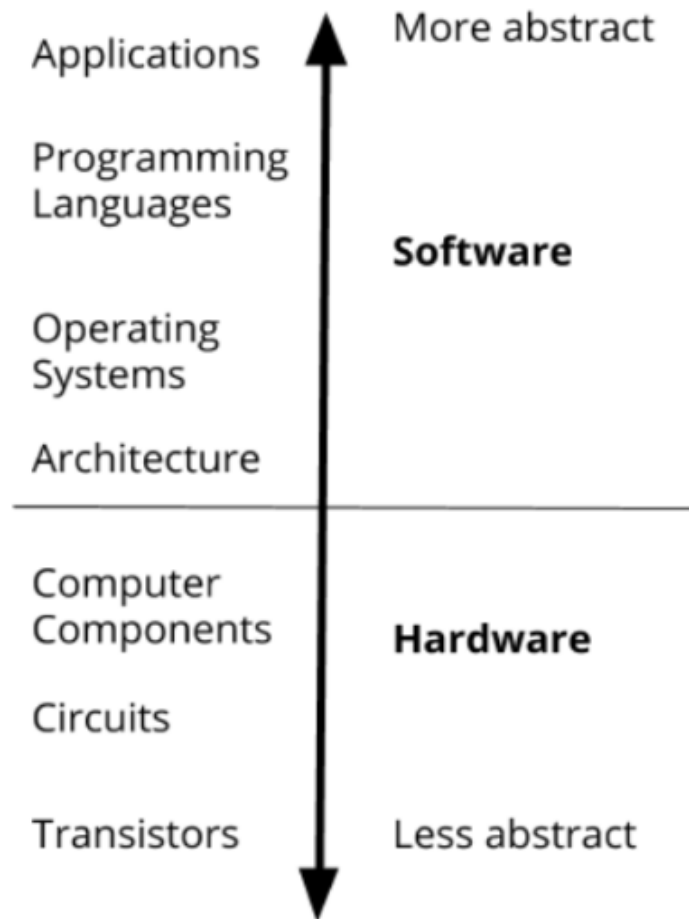


Parts of Computer System

- Computer system has four parts . This parts combindly create computer system.

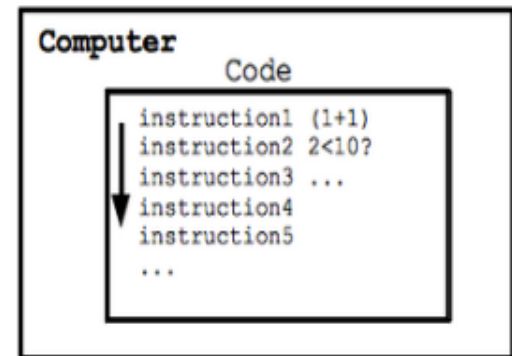


Introduction to Computers

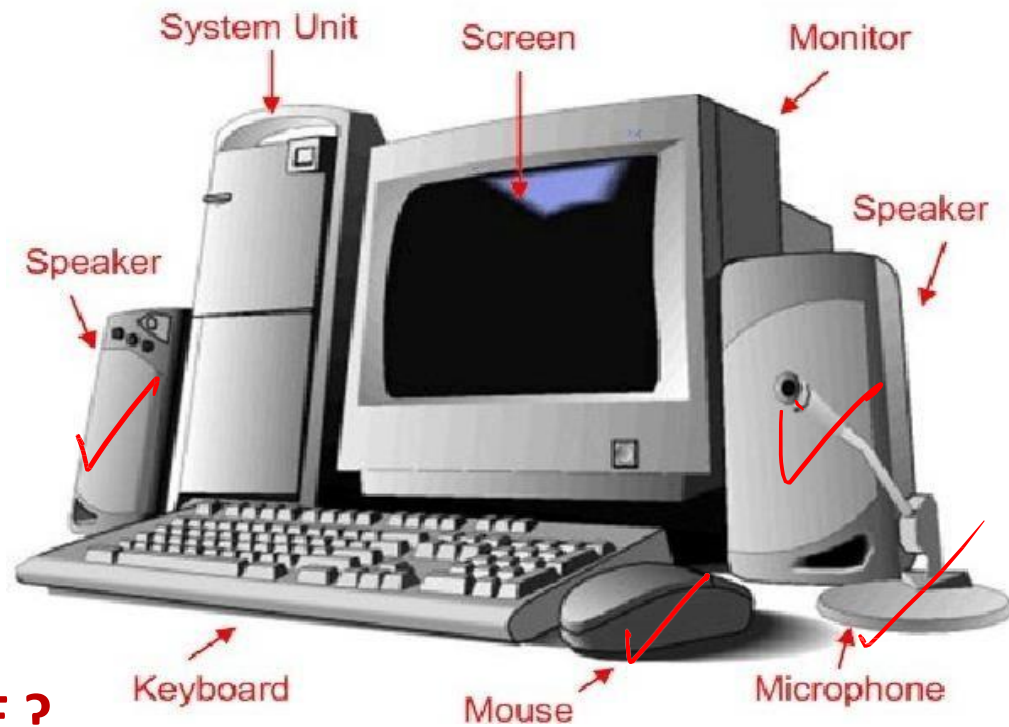


Introduction to Code: How Computers Work

- Code is a series of instructions
- Each instruction is an operation
- Computer follows each instruction, one after another
- Programs are comprised of millions of instructions



- **Physical equipment** that performs basic function of data processing.
- We can **see or touch** the parts of hardware.
- It consists of ---
 - CPU (Center processing Unit)
 - Memory unit
 - Input devices
 - Output devices
 - Secondary storage devices



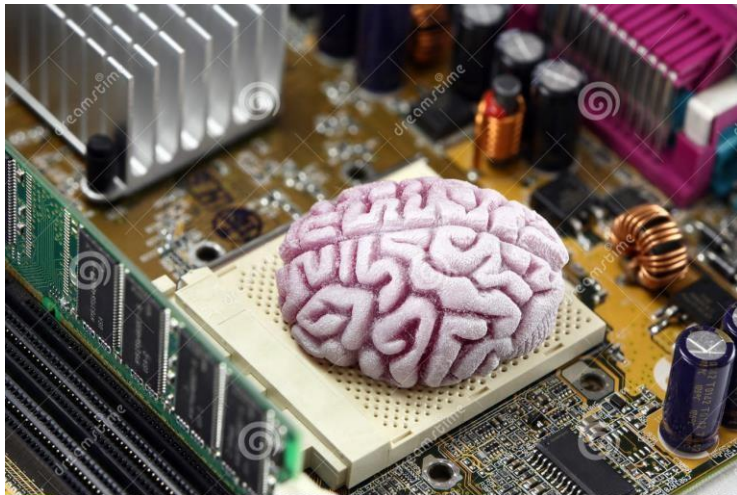
➤ **But where is the SOFTWARE ?**

CPU – Central Processing Unit

Hardware

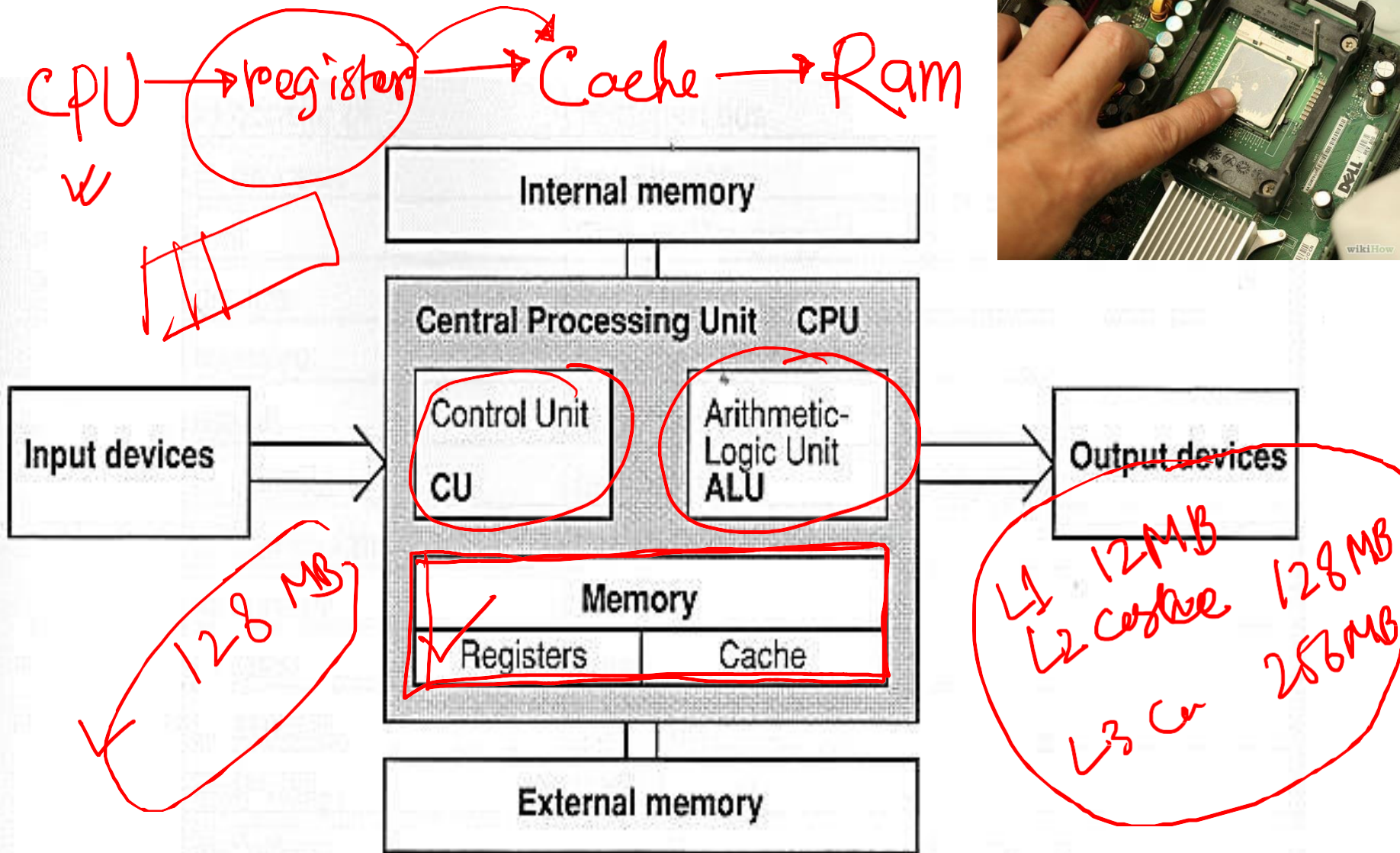
CPU consists of the following **features**:

- ✓ CPU is considered as the brain of the computer.
- ✓ CPU performs all types of data processing operations.
- ✓ It stores data, intermediate results and instructions(program).
- ✓ It controls the operation of all parts of computer.
- ✓ Processor speed 2.4 GHz = capability to execute 2.4 billion operations per second



CPU

Hardware



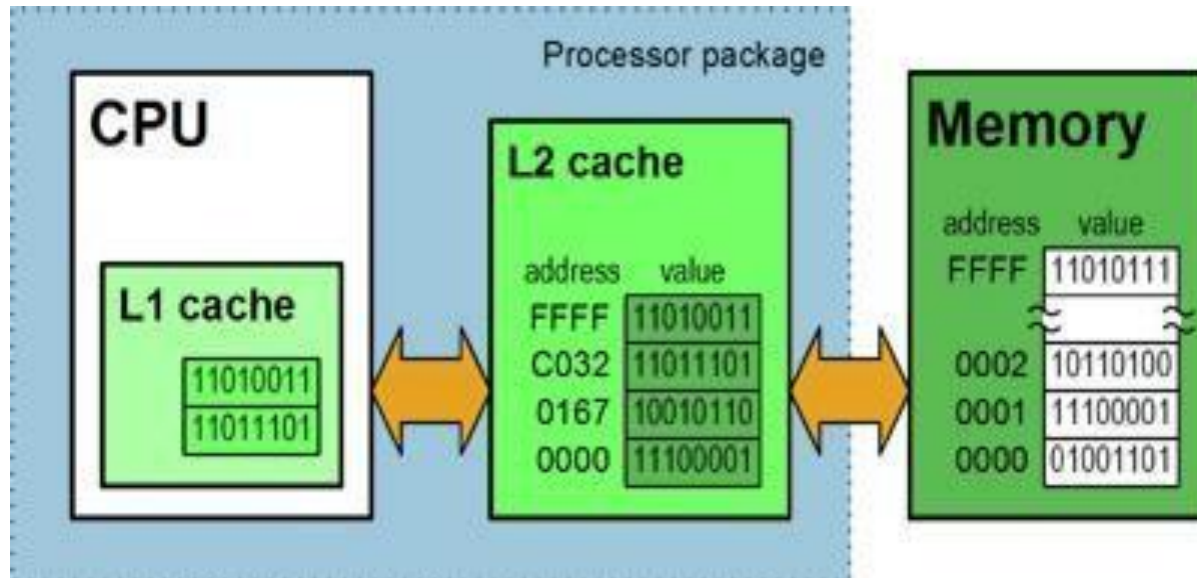
- ❖ Memory is used to store data and instructions
- ❖ Computer memory is the storage space in computer where data is to be processed and instructions required for processing are stored

Memory is primarily of **three types**

- Cache Memory
- Primary Memory/Main Memory
- Secondary Memory

Cache Memory

- ❑ Cache memory is a very high speed semiconductor memory which can speed up CPU.
- ❑ It acts as a buffer between the CPU and main memory
- ❑ supplementary memory system that **temporarily stores frequently used instructions and data for quicker processing** by the central processing unit (CPU) of a computer.



Primary Memory (Main Memory)

- ❑ Primary memory holds only those data and instructions on which computer is currently working
- ❑ It is divided into two subcategories **RAM** and **ROM**



RAM (Random Access Memory)

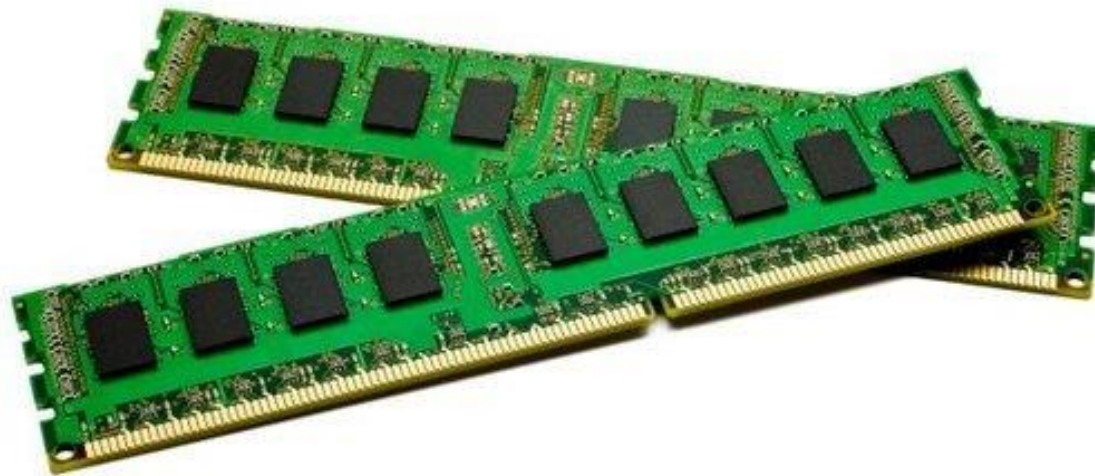
- RAM(Random Access Memory) is the internal memory of the CPU for storing data, program and program result.
- Volatile : Data is lost when powered off
- **Non - Persistent"**
 - RAM is not persistent. State is gone when power turned off
 - e.g. You're working on a doc, then power goes out and you lose your work (vs. "Save")

Small capacity

Faster access to data

- It has two types—
 - Static RAM
 - Dynamic RAM

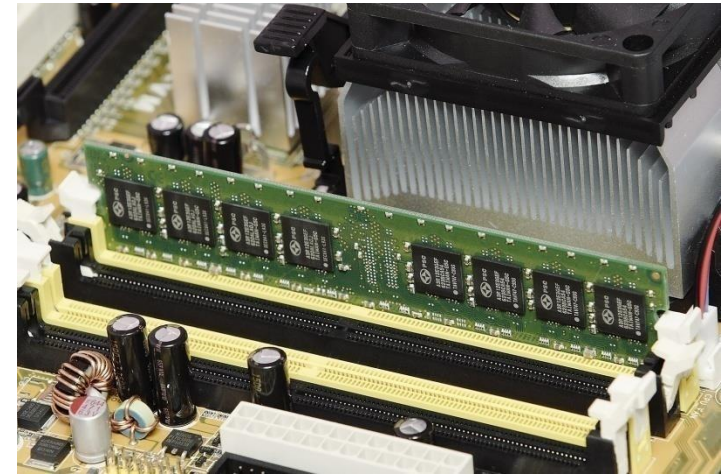
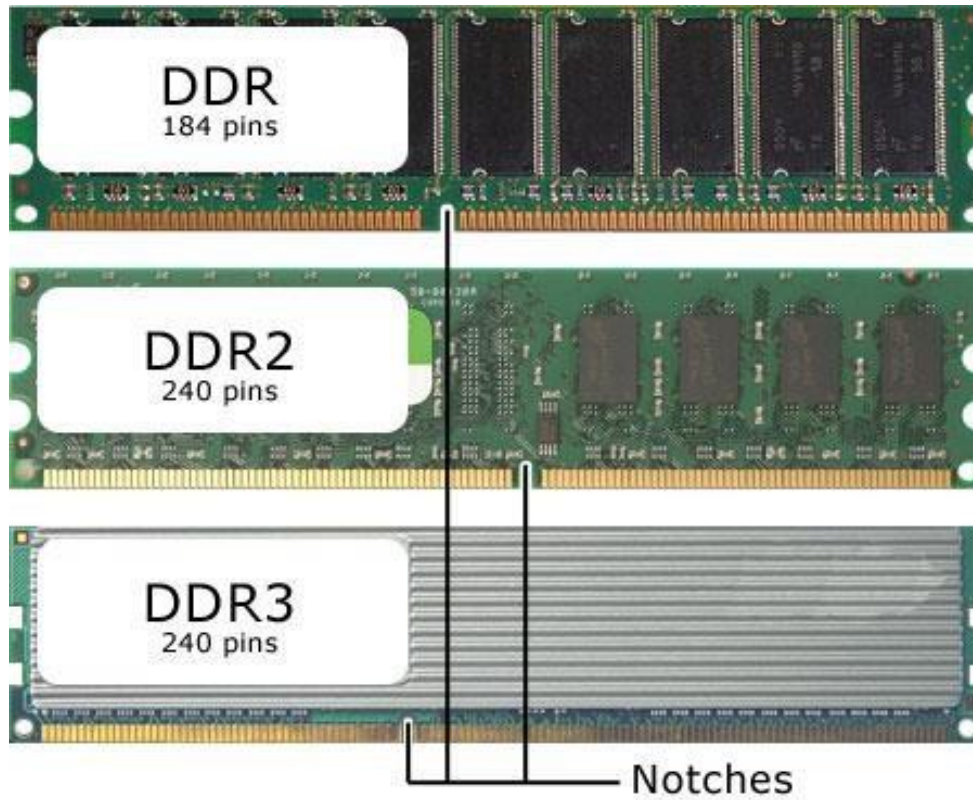




Memory - RAM

Hardware

RAM (Random Access Memory)



Give a Reading on how RAM works

Instructions are typically not stored in RAM (Random Access Memory). RAM is primarily used for temporary data storage during the execution of programs or processes. It serves as a working memory where the CPU (Central Processing Unit) can quickly read from and write to data as needed for the active tasks.

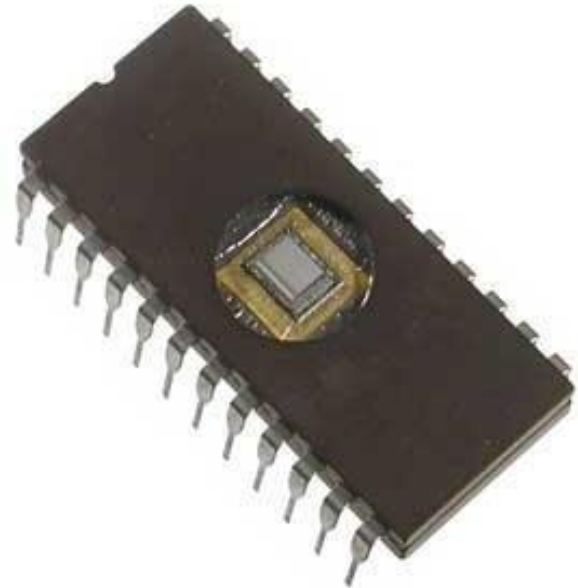
1.Program Instructions: Program instructions are typically stored in other types of memory, such as Read-Only Memory (ROM) or storage devices like hard drives or solid-state drives. When you launch a program, the instructions and data required for that program are loaded from storage (e.g., a hard drive) into RAM for execution.

2.Data: RAM is used to store data that is actively being processed or manipulated by the CPU. This data includes variables, program state, and other information required for the current task.

In summary, RAM is not meant for storing program instructions in the long term; instead, it provides a fast, volatile workspace where data and program instructions are temporarily stored while the computer or device is powered on and operational. Instructions are fetched from non-volatile memory sources (e.g., ROM or storage) and loaded into RAM when needed for execution.

Read Only Memory (ROM)

- The memory from which we can only read but cannot write on it.
- Non-volatile.
- **Information is stored permanently during manufacture**



Give a Reading on ROM

Read-Only Memory (ROM) **stores instructions and data that are permanently written during the manufacturing process and cannot be easily modified or erased by the end user.** The type of instructions stored in ROM can vary depending on the specific application and purpose of the ROM chip. Here are some common types of instructions and data that can be stored in ROM:

1.Firmware: ROM is commonly used to store firmware, which is software that is permanently programmed into a device's hardware. Firmware provides essential instructions for the device to function properly. For example, the firmware in a computer's BIOS (Basic Input/Output System) or in the ROM of a game console.

2.Bootloader: Some ROM chips store a bootloader, which is responsible for initializing the hardware and loading the operating system when a computer or device is powered on. Bootloaders are typically stored in ROM to ensure their stability and security.

3.Embedded Systems: ROM is used in many embedded systems to store the control software or firmware that operates appliances, consumer electronics, industrial equipment, and more. These instructions are permanently stored to ensure consistent and reliable operation.

Give a Reading on ROM

4. Read-Only Memory for Game Consoles: In older video game consoles, such as the Nintendo Entertainment System (NES), game cartridges contained ROM chips with the game's instructions and data. These instructions were read-only and could not be altered by the user.

5. Character Sets and Fonts: ROM can also store character sets, fonts, and other graphical or text data that needs to be readily accessible for display purposes, such as in early computer systems and printers.

6. Operating System Routines: Some ROM chips contain low-level routines and functions used by the operating system or software applications to interact with hardware devices or perform essential system tasks.

7. Security Codes and Encryption Keys: In some secure systems, ROM may store encryption keys or security codes that should not be modified or tampered with by unauthorized users.

It's important to note that modern computing devices, such as smartphones and computers, typically use a combination of different types of memory, including both ROM and RAM (Random Access Memory), to store and execute instructions and data. While ROM is non-volatile and retains its contents even when the power is turned off, RAM is volatile and is used for temporary data storage during the operation of a device.

TERMS	RAM	ROM
Definition	RAM can be defined as a temporary memory that can hold the data and instructions if there is adequate power supply.	ROM can be defined as a permanent memory holding the data even the power is switched off.
Type	The content in RAM (Random Access Memory) can be accessed and processed.	The content in ROM (Read Only Memory) can not be processed. It can only be read.
Utility	It stores immediate instructions required by the processor.	It keeps the booting instructions of a computer.
Speed & Cost	High-speed at a higher cost than ROM	Low-speed at a lesser cost than RAM

RAM	ROM
1. Temporary Storage.	1. Permanent storage.
2. Store data in MBs.	2. Store data in GBs.
3. Volatile.	3. Non-volatile.
4.Used in normal operations.	4. Used for startup process of computer.
5. Writing data is faster.	5. Writing data is slower.

Difference between RAM and ROM

Secondary Memory

- Also known as external memory or non-volatile.
- Slower than main memory.
- Used for storing data/information permanently

Floppy Drive



Secondary Memory

Hard Disk or Drive



INTERNAL STORAGE: HARD DISK AND SSD



Hard disk

- Data is stored in an array of magnetic disks
- Disks need to be rotated for data read/write

SSD: **S**olid **S**tate **D**rive

- Data is stored in an array of flash memories
- No mechanical components

Stores programs and data

Non-volatile: Data is retained even if powered off

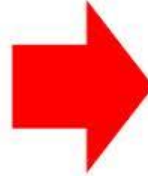
Large capacity

Slower access to data

PERSISTENT STORAGE



HDD



SSD

Persistent storage of bytes

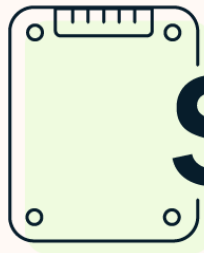
"Persistent" means preserved even when not powered

Hard Disk Drive or Hard Disk

- Made of rigid materials unlike floppy disks
- Holds a greater amount of data
 - 10MB in 1980s
 - 600MB in mid 1990s
 - 4.3GB in 1999
 - 180GB in 2001
 - 400GB - 2004



- Hard disk is a higher capacity drive.
- It also stores the operating system which runs when we turn on the computer.
- Non-volatile
- Stores data in magnetic disk
 - Speed: very fast, measured in milliseconds.
 - The smaller this number faster the disk.
 - Capacity: enormous.
 - Cost: cheap



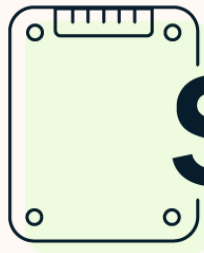
SSD

vs

HDD



faster	✓	✗	slower
shorter lifespan	✗	✓	longer lifespan
more expensive	✗	✓	cheaper
non-mechanical (flash)	✓	✗	mechanical (moving parts)
shock-resistant	✓	✗	fragile
best for storing operating systems, gaming apps, and frequently used files		best for storing extra data, such as movies, photos, and documents	



SSD

vs

HDD



faster	✓	✗	slower
shorter lifespan	✗	✓	longer lifespan
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best for storing operating systems, gaming apps, and frequently used files			best for storing extra data, such as movies, photos, and documents

Optical Drives

- | | |
|--------------|-----------------------------------|
| ■ CD-ROM | read CDs |
| ■ CD-Writer | read/write CDs |
| ■ DVD-Combo | read/write CDs, read DVD |
| ■ DVD Writer | read/write CDs
read/write DVDs |

Secondary Memory

Optical Drive.. CD, DVD



Secondary Memory

Flush/Pen Drive

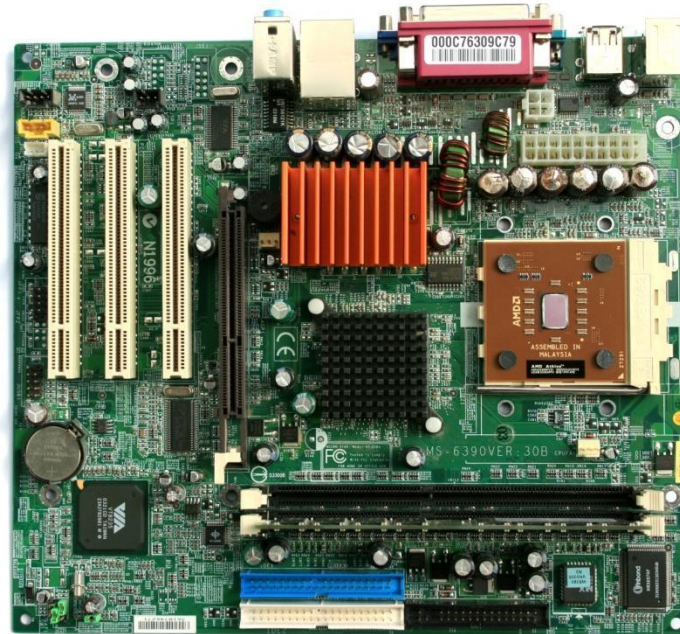


i-FlashDriveHD

The only two way storage device
between iOS and Mac / PC



- ❑ Serves as a single platform *to connect all of the parts of a computer* together.
- ❑ Connects CPU, memory, hard drives, optical drives, video card, sound card, and other ports and expansion cards directly or via cables.
- ❑ It can be considered as the **backbone of a computer**.
- ❑ **Provides data path** between components



PCI SLOTS

- **PCI slots** This board has 4 slots for the newer PCI boards for peripherals like video cards, sound cards, internal modems, etc



MEMORY SLOT

- **Memory slots**
There are 4 short slots for SIMM memory.
This board has two long slots for a new kind of memory called DIMM DRAM.



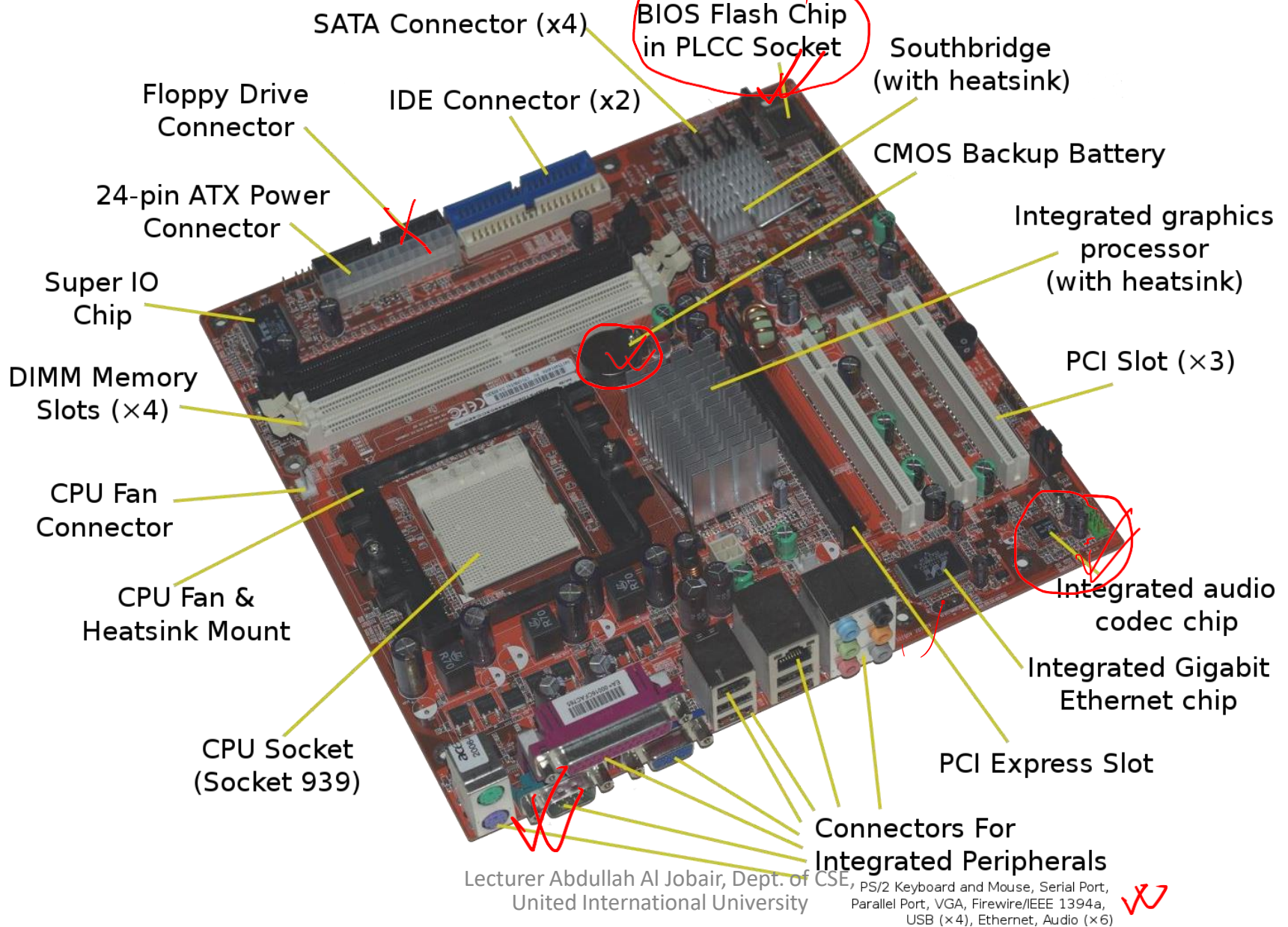
Keyboard plug : This is where the keyboard attaches, thru the back of the computer.

Power Connections This is where the power supply connects to the motherboard

Processor socket What is visible here is the place where the processor plugs into the motherboard. When the processor is installed, you still can't see it because on top of it is a heat sink and fan to keep the processor cool. Hot processors make mistakes or even melt important parts on the chip. Different processors are different sizes. So the socket on the motherboard has to match the processor. Also, the circuits in the motherboard itself must be different for different processors.

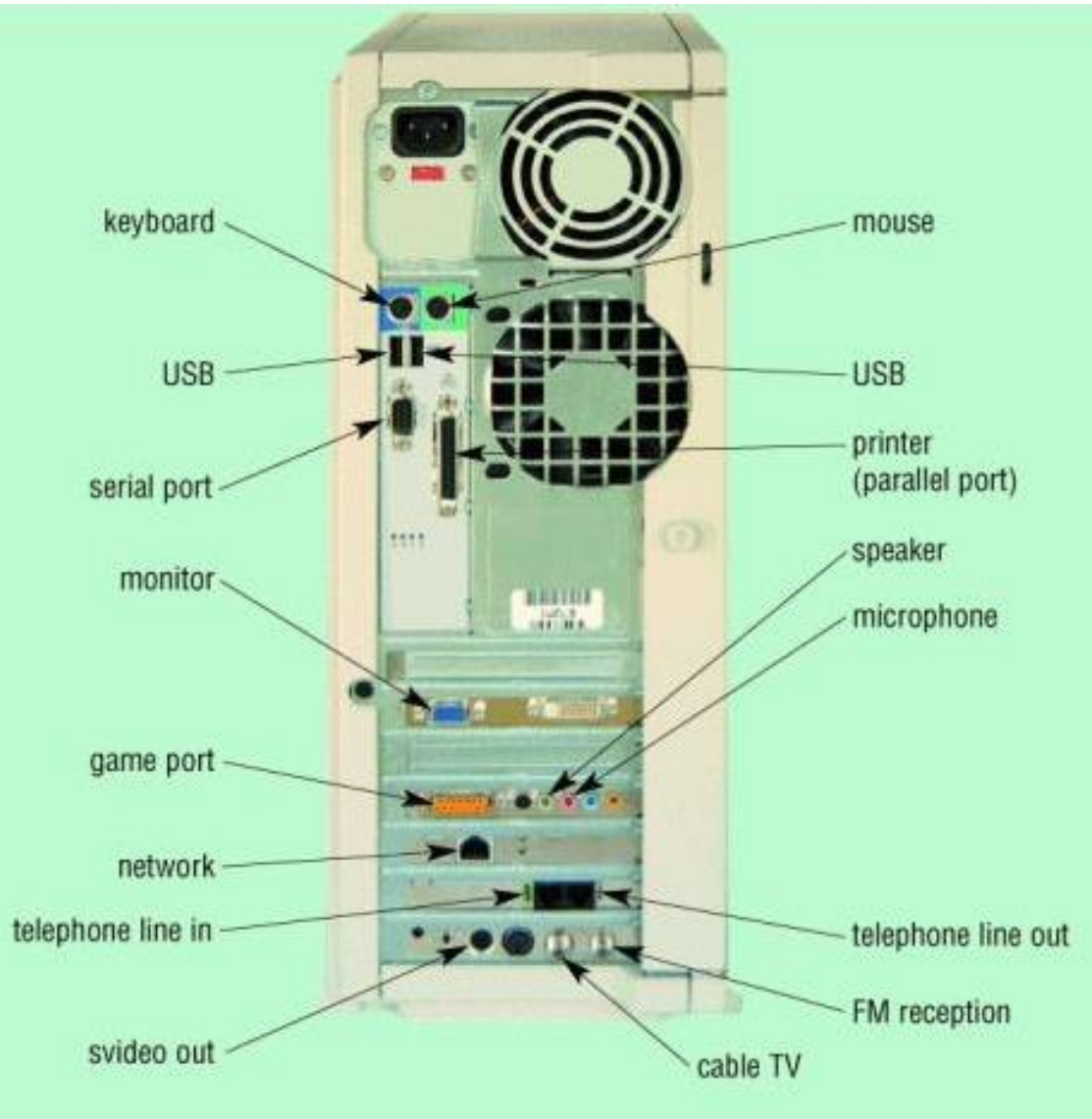
Motherboard

Ports



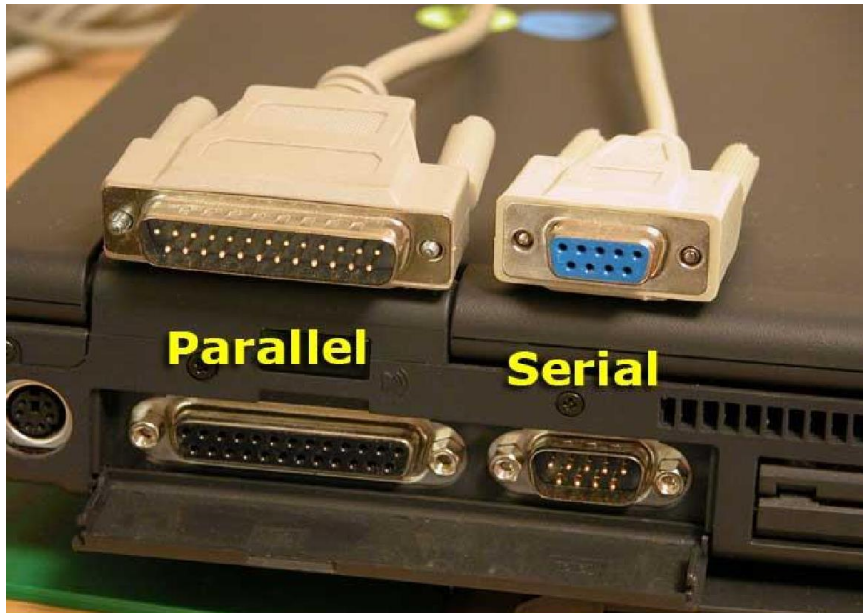
Motherboard

Ports



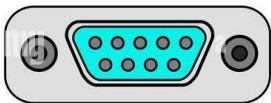
- Ports are sockets that allow you to plug in device connectors to access the common electrical bus on the motherboard.
- Ports are usually found on the back of the system unit, but newer styles also have some of them conveniently located on the front.
- Ports allow specific types of connectors (which partly reflects changing technology as well as various kinds of technology).

Types of Ports



Serial Port

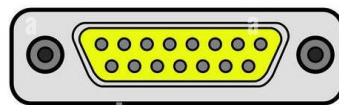
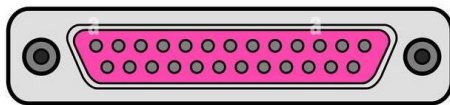
PS/2 Port



Parallel Port



Games Port



All replaced by USB

- Serial ports transmit data one bit at a time, like the picture on the left illustrates.
- Parallel ports transmit more than one byte at a time.
- These types of port designs are based on whether or not fast data transmission rates are required by the device or not.
- Most computers come with basic types of ports (serial, parallel, keyboard, mouse, and USB); and expansion cards allow you to expand the available types needed by specific devices.

- Expansion slots are sockets to provide direct connections to the common electrical bus, allowing you to insert a circuit board into the motherboard.
- Typical Expansion Cards:
 - Video Cards
 - Sound Cards
 - Modem Cards
 - Network Interface Cards (NIC)
- Laptops and portable computers typically have PC Cards – thin credit-card sized devices used to add memory, disk drives, etc.



- Allows data and programs to be sent to the CPU.

- ☐ Keyboard
- ☐ Mouse
- ☐ Joystick
- ☐ Microphone
- ☐ Webcam
- ☐ Scanner
- ☐ Monitor

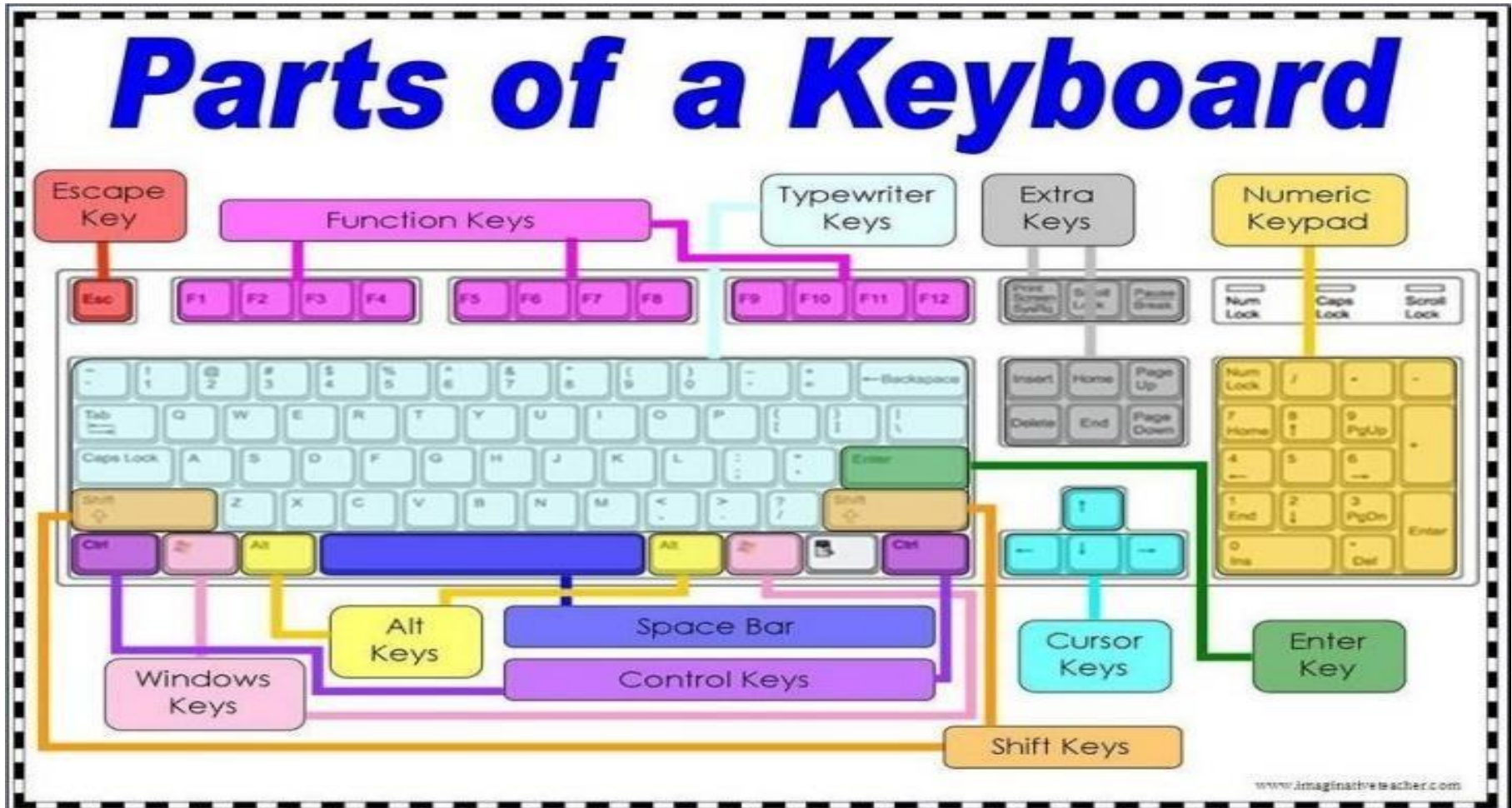


Keyboard

- Traditional keyboards
- Flexible keyboards
- Ergonomic keyboards
- Wireless keyboards
- PDA keyboards



Keyboard



Mouse

Two Types of Mouse

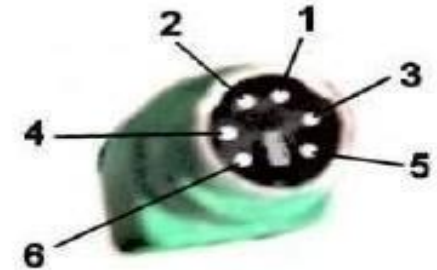
- **Mechanical** - a type of computer mouse that has a rubber or metal ball on its underside and it can roll in every direction.
- **Optical:** This type uses a laser for detecting the mouse's movement.



Mouse

How a Mouse Hooks Up to a PC

- PS/2 Mouse

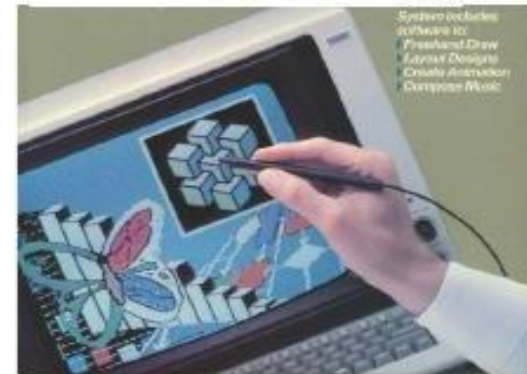


- Serial Mouse

- USB/Cordless Mouse



- Joystick – input device for computer games
- Light Pens – light-sensitive penlike device
- Stylus – penlike device commonly used with tablet PCs and PDAs.



Monitor

Types of Monitor

- Cathode Ray Tube (CRT)
- Liquid Crystal Display (LCD)



Printers

- **IMPACT PRINTERS** uses pressure by physically striking the paper. Ex. Daisy wheel printers, line printers, dot matrix printers & band printers.
- **NON-IMPACT PRINTER** does not apply pressure on the paper but instead produces character by using lasers, ink spray, photography or heat.

Output Devices

Hardware



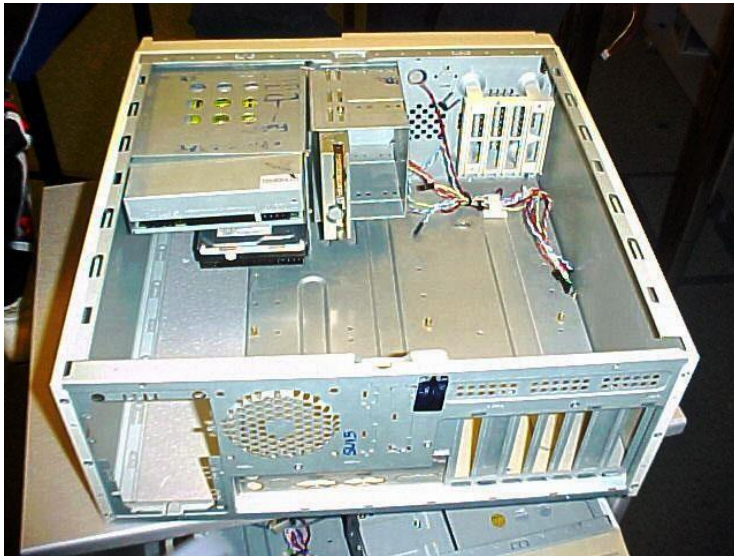
Dot matrix
printer



Laser
printer



Inkjet printer



Casing or cover

- The box or outer shell that houses most of the computer, it is usually one of the most overlooked parts of the PC.
- Protects the computer circuits, cooling and system organization.

Power Supply

- Responsible for powering every device in your computer.
- Parts of a Power supply:
 - ☐ Disk drive connectors
 - ☐ Motherboard connector
 - ☐ Power supply fan
 - ☐ Power switch
 - ☐ Input voltage selector
 - ☐ Cover
 - ☐ Power plugs receptacle



SOFTWARE

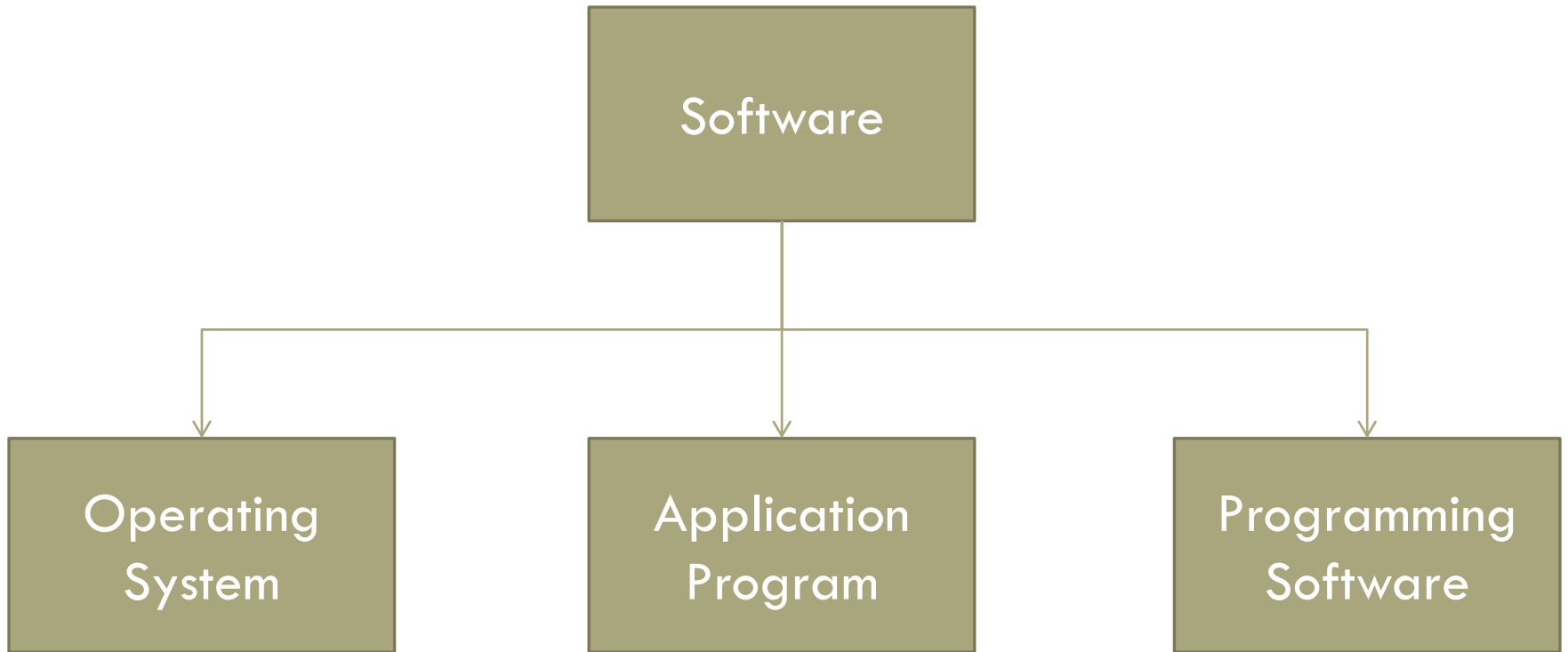
WHAT IS SOFTWARE?

A set of instructions that tells a computer to do a specific task

Software tells the computer where to take input from, how to process it, and how to present the result to the user

All these things happen behind the scene (abstraction)

THREE TYPES OF SOFTWARE



OPERATING SYSTEM



Manages hardware and software resources

- Communicates with the hardware
- Allocates memory to software
- Provides common services for programs (e.g. network connectivity, peripherals etc.)

Example: Windows, Linux, MacOS, iOS, Android

APPLICATION PROGRAM



Any program that is suited for real life tasks (browsing, writing, playing games etc.)

- Google Chrome, Microsoft Word, Microsoft PowerPoint

.exe (Windows) or .app (Mac) file

Runs on top of operating systems

- Depends on operating system for resources
- **That is why you cannot run a .exe file in Mac, or a .app file in Windows**
 - Can use without knowing how it works behind the scenes (**abstraction**)
 - Programs are written in **code**, the next layer of abstraction

PROGRAMMING SOFTWARE

Helps write, debug and execute programs, thus creating software

Useful for developers, but not used by normal users

Example: GCC, Java, PHP, Python, Visual Studio, Codeblocks, Netbeans, Android Studio